MCM2025-D 真题解析

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目 录
CONTENT

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PART ONE

2025年D题介绍

SJTU

2025年D题介绍

- 美赛D题是**运筹学与网络科学类**建模问题。通常要求运用多目标优化、系统动力并结合算法优化解决复杂系统问题。

- 本题要求参赛团队以美国巴尔的摩市为研究对象，为该市设计一套兼顾多方需求的**交通网络优化方案**，解决基础设施老化、桥梁坍塌等各种情况带来的交通困境，通过优化城市交通网络打通**经济**联通壁垒。

2025年D题介绍

- **任务一：分析桥梁倒塌或重建桥梁的影响**
  - 核心任务：为巴尔的摩交通系统建立网络模型
  - 重点说明对巴尔的摩市内及周边各利益相关者的影响

如：出行居民、交通运输机构、政府

- **任务二：分析（潜在）项目对交通系统的影响**
  - 核心任务：寻找合适的分析项目
  - 重点说明对巴尔的摩市内及周边各利益相关者的影响

注意点：题目描述部分额外提及了城市经济发展和绿色环保需求，分析项目影响时不能局限于通勤

- **任务三：推荐一个最能改善巴尔的摩居民生活的项目**
  - 项目对居民的好处
  - 项目对其他利益相关者的影响
  - 项目对其他交通需求和人们生活的干扰

缩短通勤时间、环境保护、经济发展

2025年D题介绍

- **数据文件:**
  - **Bus_Routes.csv**：2022年巴尔的摩市MTA公交路线
  - **Bus_Stops.csv**：2022年巴尔的摩市MTA公交站位置
  - **nodes_all.csv**：由 OpenStreetMaps 标记的道路节点，一般是两条交通路径相交的位置
  - **nodes_drive.csv**：由 OpenStreetMaps 标记的汽车行驶道路节点，一般是两条道路或高速公路相交的位置
  - **edges_all.csv**：nodes_all.csv 数据集中两个节点之间的道路
  - **edges_drive.csv**：nodes_drive.csv 数据集中两个节点之间的车道
  - **Edge_Names_With_Nodes.csv**：将 nodes_all.csv 数据集的信息与 edges_all.csv 数据集的信息配对，提供带有节点的道路

2025年D题介绍

- **数据文件：**
  - **MDOT_SHA_Annual_Average_Daily_Traffic_Baltimore.csv**：马里兰州交通部州公路管理局年平均日交通量（AADT）数据，由线性和点几何特征组成，代表整个马里兰州包含交通量信息的地理位置和路段。交通量信息根据交通计数生成，用于计算年日平均交通量 （AADT）、年平均工作日交通量 （AAWDT） 以及基于车辆类别年平均日交通量
  - **DataDictionary.csv**：解释了上述所有文件中变量名的含义。

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PART TWO

思路分析

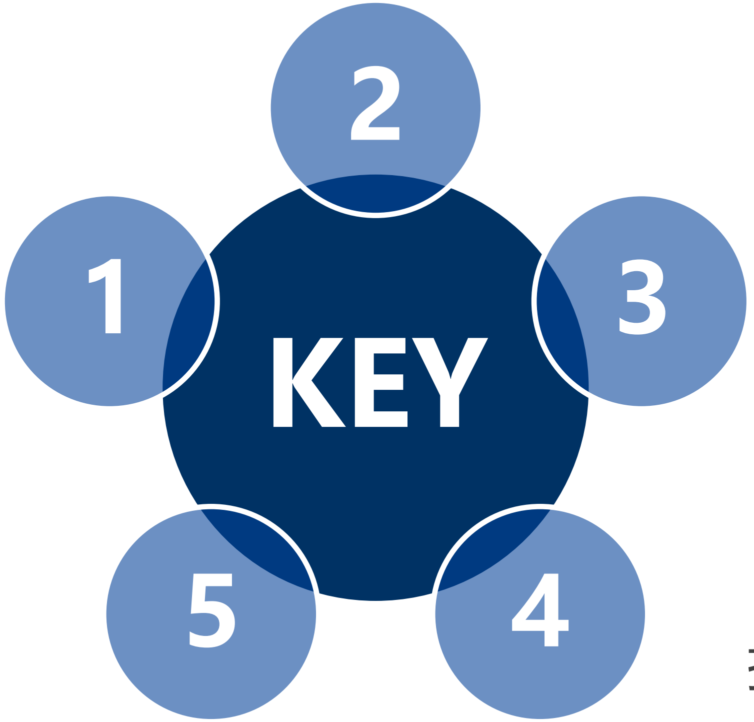
框架总览

Task 1：

突发事件影响评估（桥梁坍塌）

基础构建：

识别问题、数据处理、构建网络



Task 2：

选择已有/潜在交通项目

利用模型评估多方影响

总结：

报告与备忘录

Task 3：

推荐最能改善居民生活的项目与全面评估

基础构建：识别问题、数据处理、构建网络

- **问题与目标细化**：
  - **核心问题**：巴尔的摩交通基础设施老化；交通选择有限；弗朗西斯·斯科特·基桥坍塌
  - **终极目标**：提高居民生活质量；优化交通网络；促进经济发展；交通安全；环境可持续……
  - **利益相关者（Stakeholders）分类**：明确六类利益相关者（城市居民、郊区居民、通勤者、商家、过路旅客、游客）的需求和出行方式，作为贯穿整个项目的评估.
  - **案例分析**：题目中提供的具体交通问题（三小问），是模型验证和应用的重要场景。
- **数据获取与预处理**：
  - **数据清洗**：对提供的CSV数据进行去重、处理缺失值和异常值（如采用**3σ原则**）。
  - **数据预测**：利用历史交通流量数据（AADT: annual average daily traffic），结合合适的预测模型（如**Verhulst模型**、时间序列分析），预测未来基线交通流量。

基础构建：识别问题、数据处理、构建网络

- **交通网络模型构建：**
  - **建图（图论）**：将巴尔的摩交通系统抽象为有向图或多层网络**(e.g. G(V, E, A, W))**。
  - **节点（Nodes）**：代表交叉口、公交站点、关键地标等。
  - **边（Edges）**：代表路段、公交线路、步行路径。
  - **多模式集成：**构建多层分区交通网络（Multilayer-Zoned Transportation Network），将驾驶、公交、步行等不同模式整合到统一框架中，并考虑层间连接。
  - **属性赋值：**为节点和边赋予实际交通和地理属性，e.g.
    - **边属性：**长度、车道数、限速……
    - **节点属性：**地理坐标、连接度……
  - **关键点高亮：**识别并高亮显示高流量交叉口、重要道路、以及城市关键地标
  - **区域划分：**将城市划分为功能区，以便进行区域性影响分析。

建图案例(F奖)

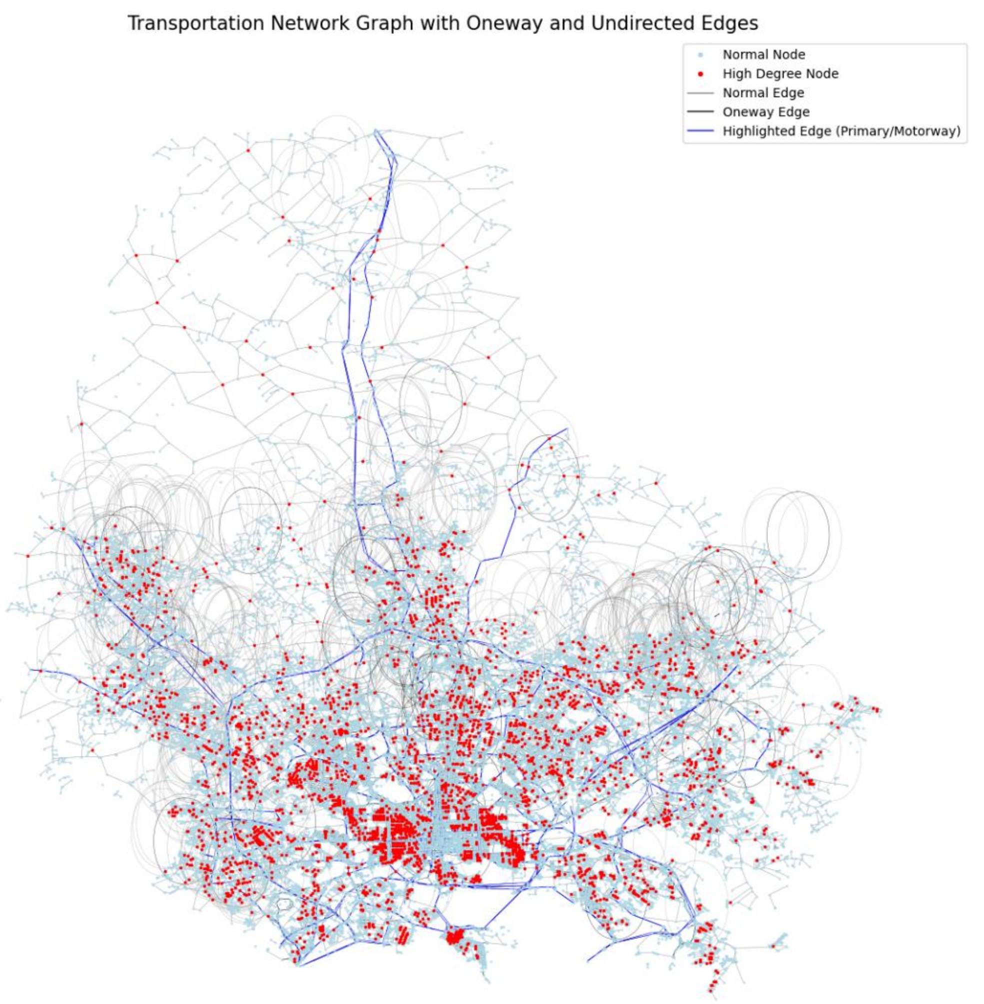


Figure 1: Basic model

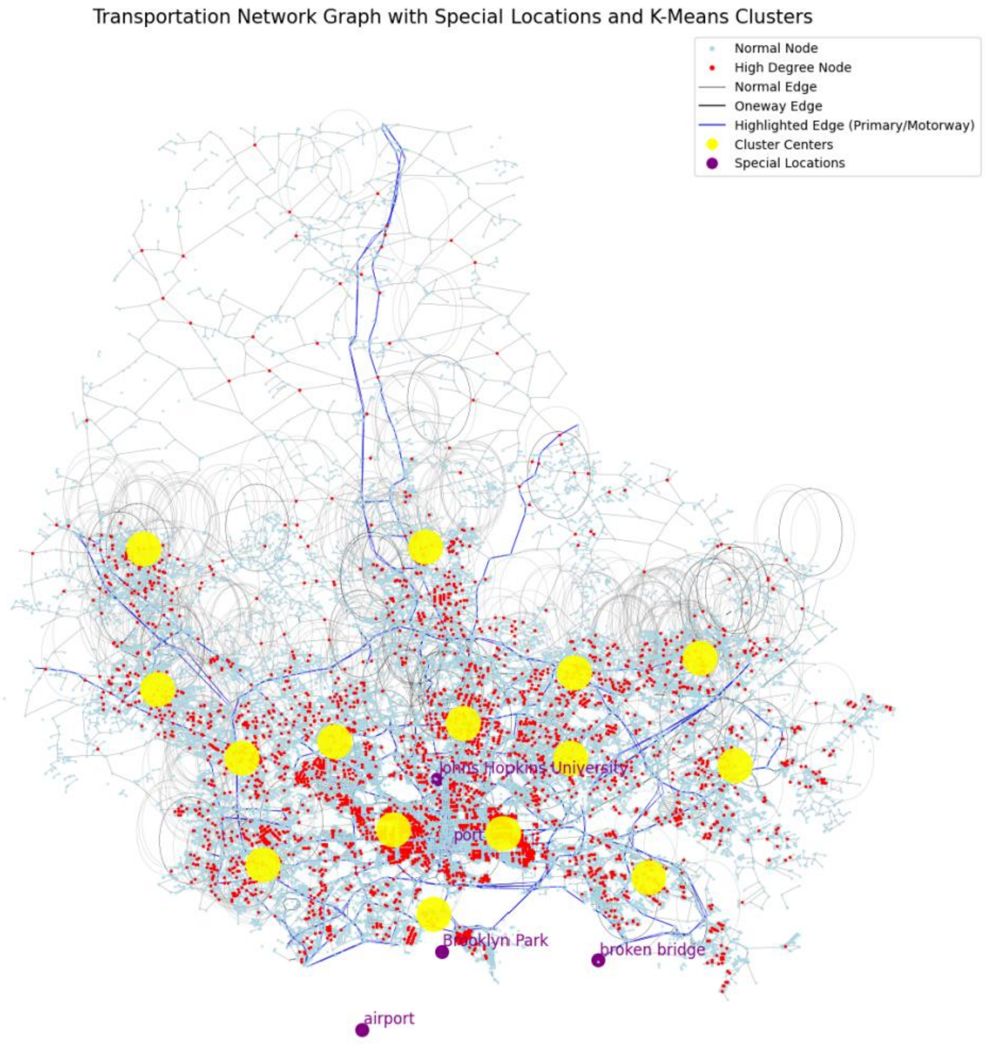


Figure 2: complete model

Task 1：突发事件影响评估（桥梁坍塌）

坍塌情景模拟与流量再分配：

1. **模型操作**：在已构建的网络中，将代表弗朗西斯·斯科特·基桥的**相关边移除**或将其**通行能力设为零**。

2. **交通流分配**：
   - ➢ 应用 **Wardrop's Equilibrium Principle**，模拟驾驶员在网络中重新选择最短路径，达到新的平衡状态。
   - ➢ 采用 **Tree-Based Traffic Allocation Model** 模拟交通流在替代路线上的扩散和分配。
   - ➢ **流量扩散分析**：模拟流量如何在整个网络中扩散，生成城市范围内的交通流图。

多维度量化影响：

1. **交通指标：**
   1. 对比坍塌前后关键路段的**交通流量**变化。
   2. 计算**平均旅行时间**和**交通饱和度**的变化。
   3. 评估**网络整体效率**的下降。

2. **利益相关者影响：**
   1. 利用分区**Monte Carlo模拟**或基于**最短路径算法**计算不同利益相关者（通勤者、商家、游客）的出行成本增加。
   2. 分析货运成本、紧急服务受阻等。

3. **区域影响**：通过**K-means聚类**将关键节点划分为不同区域分析坍塌对不同区域交通流和通勤时间的影响.

Task1 图例

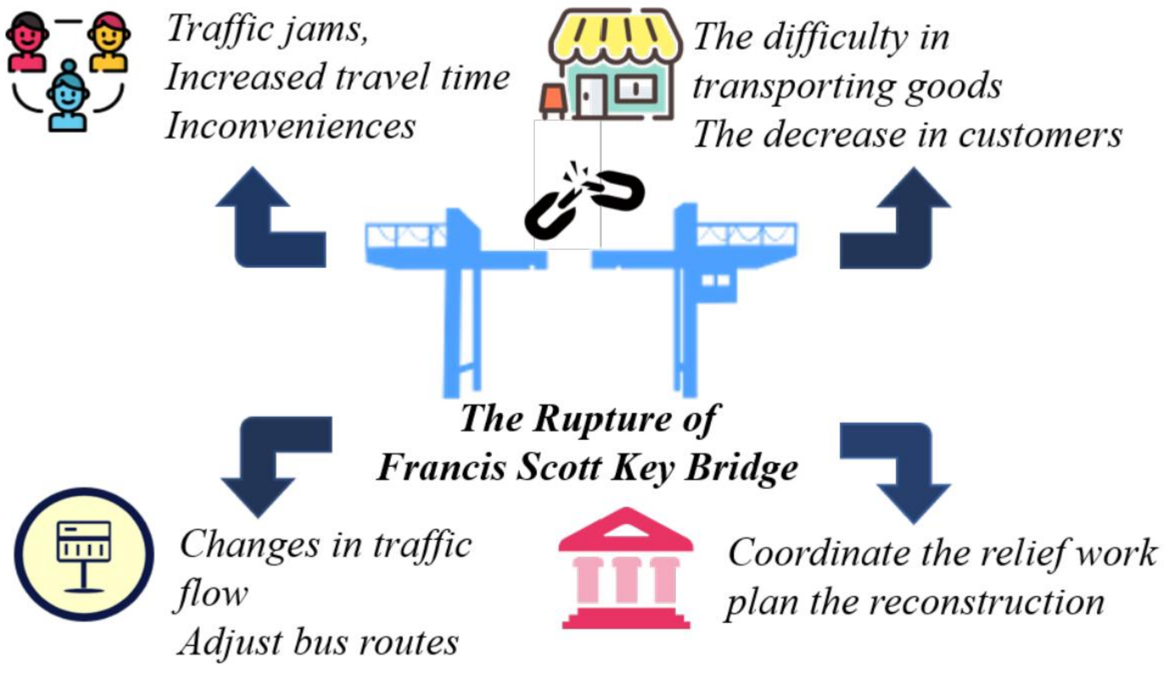


Figure 6: Impact of Francis Scott Key Bridge Collapse

O奖

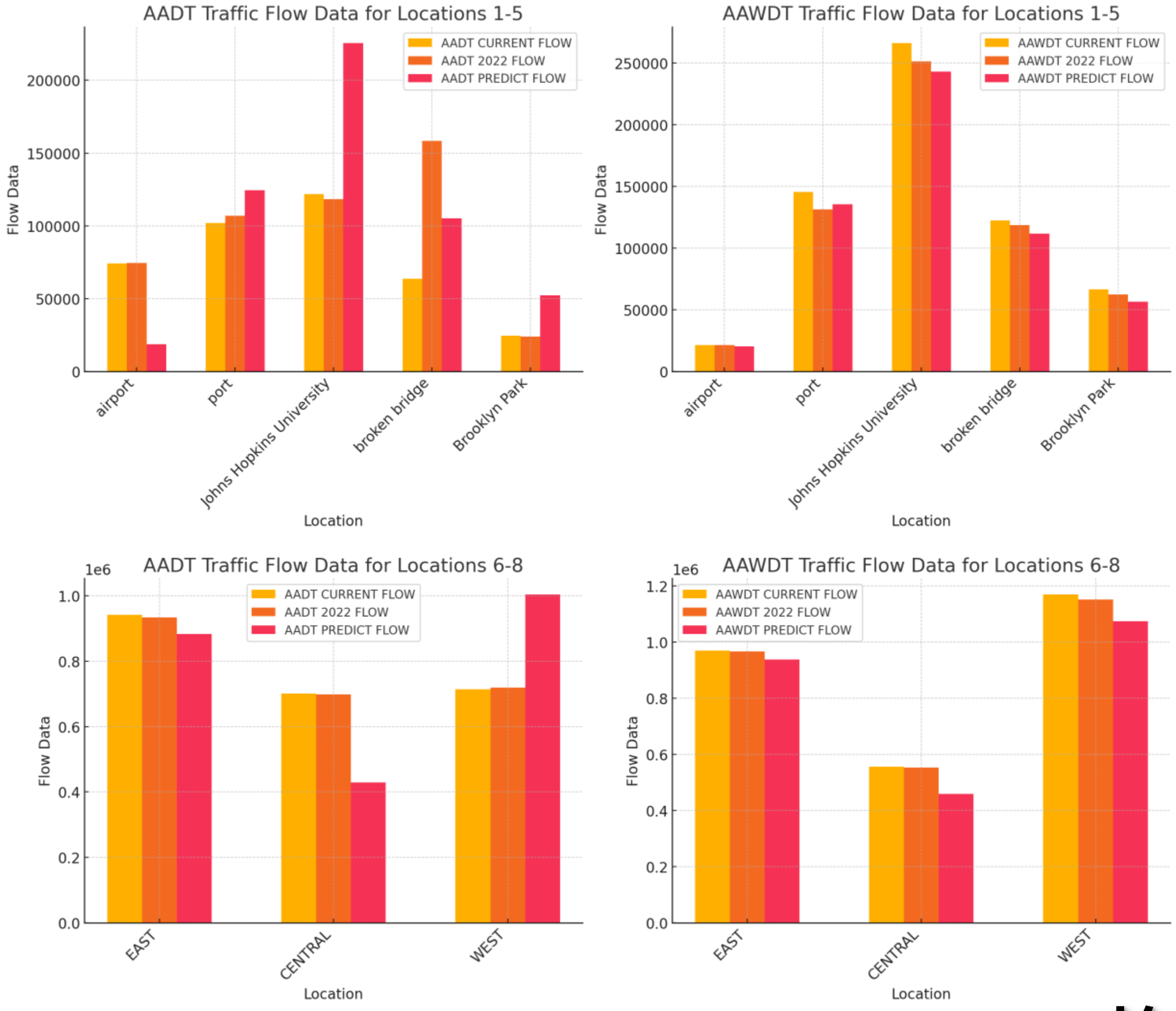


Figure 6: Bar chart of traffic flow

F奖

Task 2：利用模型评估自选项目多方影响

项目选择与建模：

1. **已有项目**：
   - ➢ 巴尔的摩公共交通系统（F奖）
2. **潜在项目**：
   - ➢ 快速公交系统（BRT）（O奖）
   - ➢ 公交专用道
   - ➢ 新增或优化公交站点
3. **网络集成**：（以公交为例）
   - ➢ **独立网络**：将巴尔的摩的公交站点和公交线路构建成一个独立的网络图。
   - ➢ **叠加到城市主网络**：将这个公交网络与已构建的城市道路网络叠加，形成一个多模式（公路+公交）交通网络，以便分析公交系统在整个城市交通体系中的作用。

Task 2：多维度评估公交系统功能与潜在项目影响

评估现有项目的价值：

1. **控制变量实验：**
   - 完整网络 VS 缺失网络 **（减法原理）**
   - 量化指标：
     - 公共服务 - 覆盖面积
     - 市民生活 - 通勤便利
     - 距离、速度、时间（**Dijkstra算法**）
2. **结论分析**：
   - 分区：市中心 VS 郊区
   - 分对象：通勤者、商家、游客……

评估潜在项目的影响：

1. **扩展模型（加法，以BRT为例）：**
   - 假设建设一条快速公交（BRT）线路
   - **多模式公交-行人网络**：构建一个包含公交、行人路径和换乘链路的多层有向图。
2. **利益相关者影响：**
   - **通勤模型**：复杂模式特定出行时间函数，考虑等待时间（Wellbull分布）、行驶时间、拥堵以及环境（PM2.5）因素。
   - **需求分配模型**：使用**嵌套Logit模型**来模拟乘客在不同交通模式间的选择偏好。
   - **可达性与公平性**指标：引入可达性潜力（Accessibility Impact Metric）和 Gini Coefficient 来量化BRT项目对区域的覆盖能力和空间公平性。

Task 2 图例

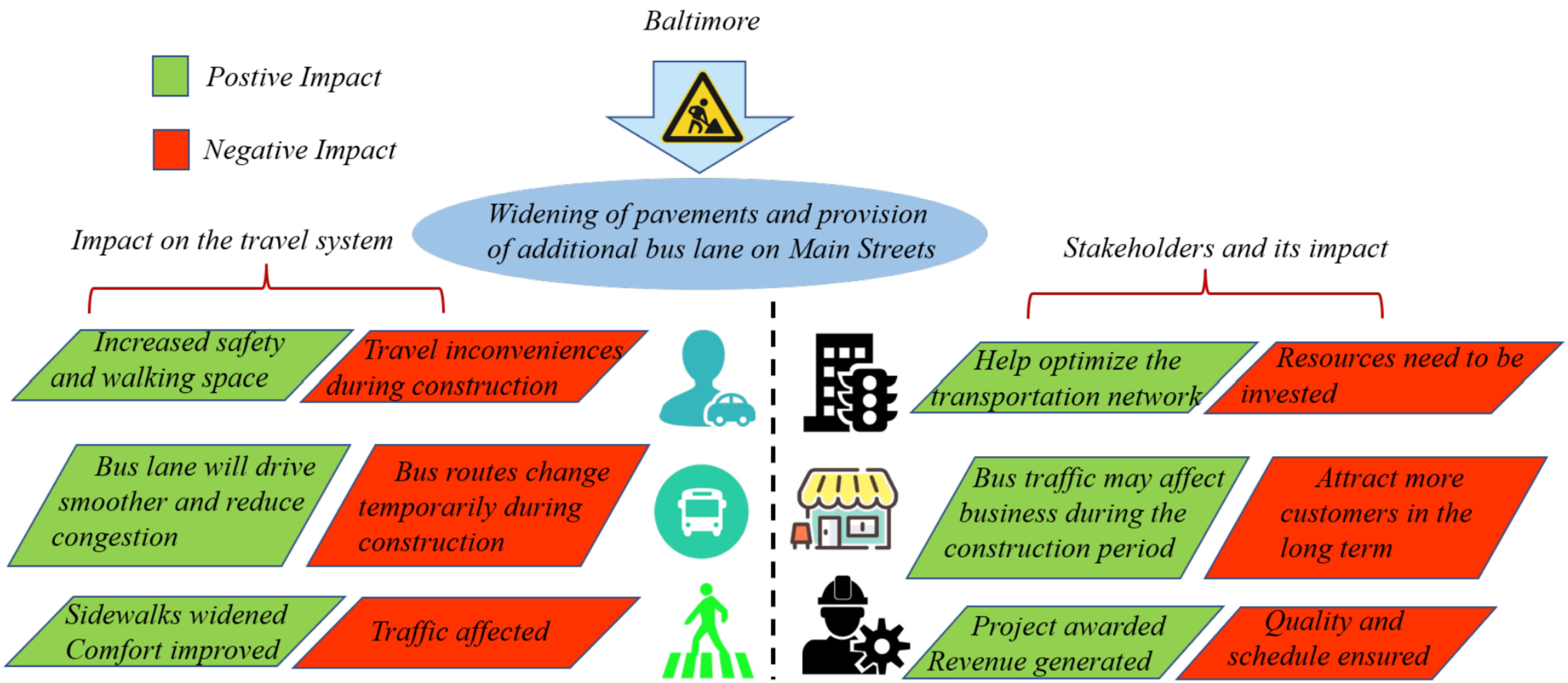


Figure 9: Impact of transport projects

Task 3：推荐最能改善居民生活的项目

核心思路：多准则决策、方案优化、全面综合评估

1. **项目选择：**
   - ➢ **多准则决策分析（MCDA）**：考虑城市交通改善涉及的多方利益和复杂目标（成本、效益、环境、公平）
   - ➢ **层次分析法（AHP）**：目标层（选择最优交通项目），准则层（如：**通勤时间**、**覆盖范围**、**成本**、**环境影响**），以及方案层（例如：重建桥梁、改善公交网络、建设地铁、增加公交站点等多个候选项目）。

2. **实施项目：**
   - ➢ 转化为交通网络中的图论问题
   - ➢ 识别目标区域
   - ➢ 设计目标函数
   - ➢ 模拟优化：
     - ➢ Dijkstra算法
     - ➢ Simulated Annealing算法

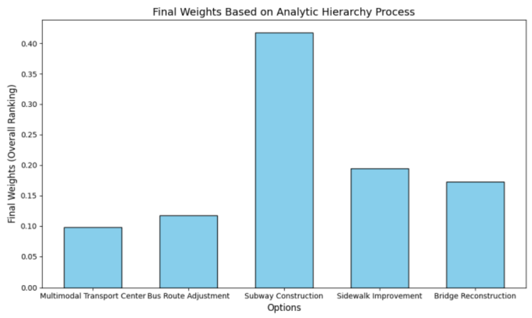


Figure 13: Results of Analytic Hierarchy Process (AHP)

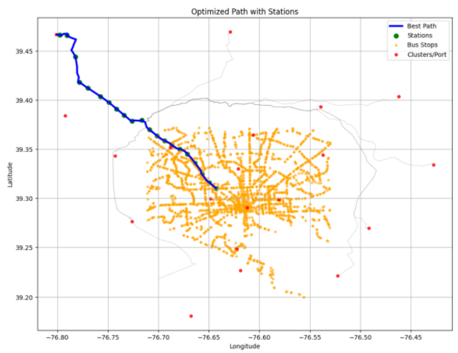


Figure 14: Optimal subway route

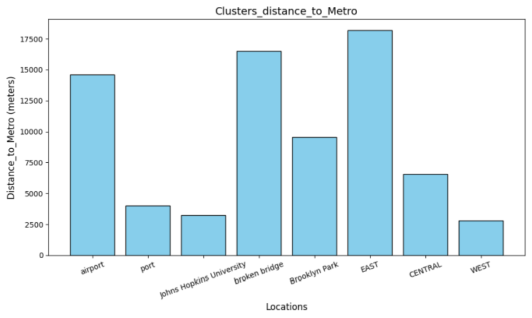


Figure 15: Distance Between People in Different Regions and Subway Line

Task 3：全面评估推荐项目

居民

通勤时间
可达性
出行效率
经济可负担性

干扰

施工影响
现有系统调整

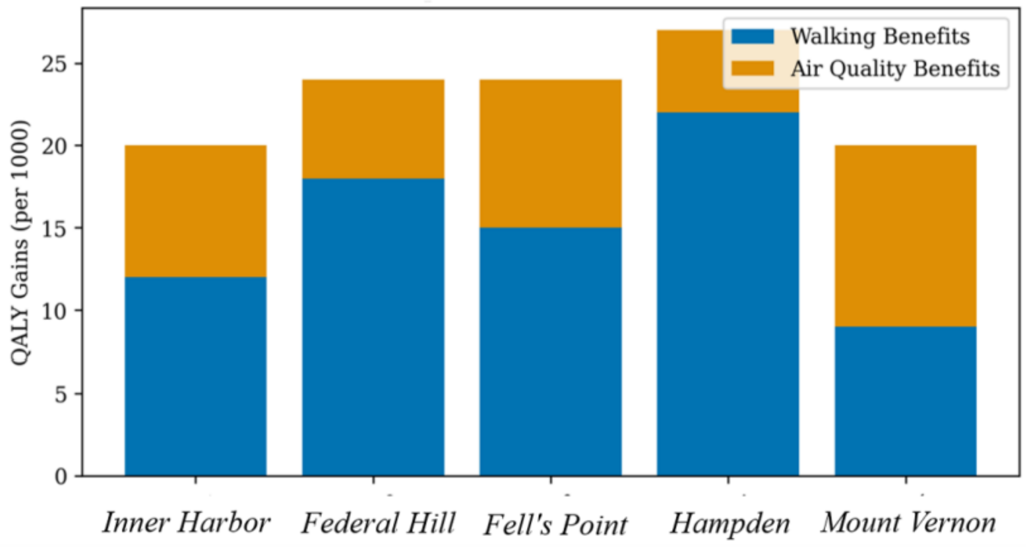
其他利益相关者

企业/物流效率
游客旅游体验
环境保护

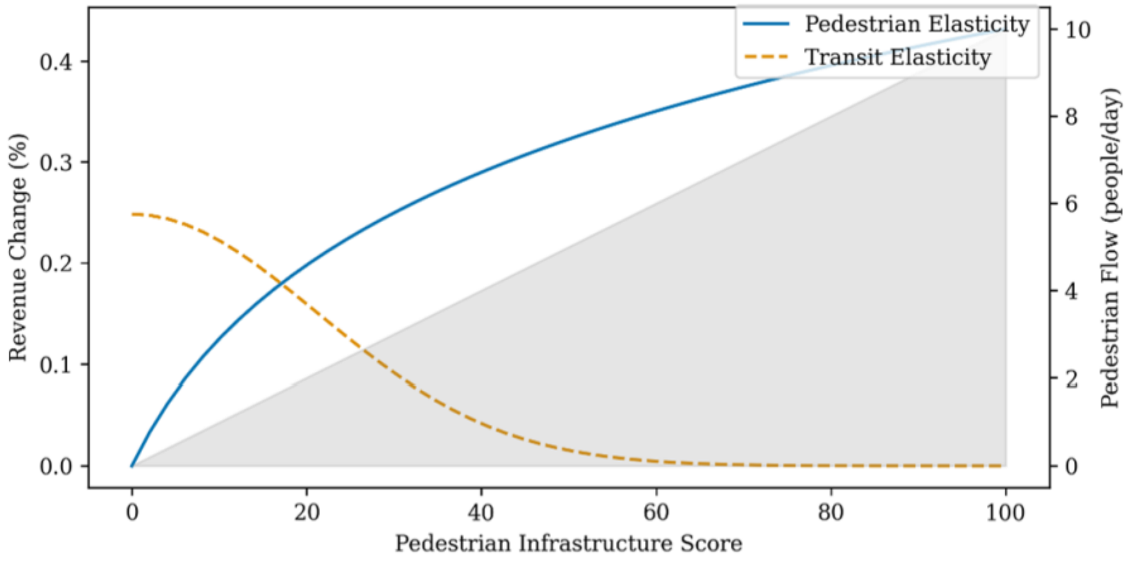
经济发展

区域经济发展
吸引投资
增加就业

Task3 评估图例



(a) Decomposition of Health Benefits



(b) Commerciasl Revenue Elasticity

Figure 11: Quantifying residents' well-being

总结：报告与备忘录

报告撰写

- **结构清晰**：背景、模型、结果、讨论、结论、优劣
- **数据可视化**：如交通网络模型、条形图、热力图、聚类分布图等

给市长的备忘录

- **读者意识**：非技术性语言、简洁明了、可操作的政策建议
- **重点突出**：强调项目的主要效益以及潜在的局限

模型评估与讨论

- **优缺点分析**：创新性、全面性、实用性模型的局限性、简化假设
- **敏感性分析**：验证模型对参数变化的鲁棒性，增强模型的可靠性

呈现最终成果

凸显模型价值

论文总评

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PART THREE

五档论文总评

A Multi-Methodological Framework for Post-Disaster Traffic Optimization and Sustainable Mobility Planning in Baltimore

Summary

Urban transportation systems face critical challenges in balancing efficiency, resilience, and sustainability, particularly under infrastructure disruptions. This study presents a multi-methodological framework to address post-disaster traffic optimization, multimodal network equity, and sustainable mobility planning in Baltimore, Maryland.

In order to address the influence of the cascading traffic disruptions caused by the collapse of the Francis Scott Key Bridge in Baltimore, a **fuzzy comprehensive evaluation model** is developed to quantify multidimensional impacts, integrating flow redistribution intensity (Q), detour burden (η), network efficiency degradation (ΔI), and environmental externalities (C_e). The framework employs **entropy-weighted aggregation** to prioritize factors dynamically based on historical incident data. Network dynamics are analyzed through **Wardrop equilibrium-based traffic assignment** and **dynamic flow conservation principles**. Results reveal a 41.7% surge in bypass traffic, 0.38 efficiency loss, and $2.1M annual health costs from increased emissions.

To create public transportation projects that help improve residents' mobility, a **multimodal network model** ($\mathcal{G}_M = (\mathcal{N}, \mathcal{E}, \mathcal{M})$) is proposed to evaluate bus-pedestrian-auto interactions. Mode-specific impedance functions incorporate stochastic waiting times, congestion penalties, and pedestrian safety risks. The framework quantifies accessibility equity using **Gini coefficients** and simulates Bus Rapid Transit (BRT) interventions, achieving a 44.1% bus mode share increase and 14.6% Gini reduction. Algorithmic integration of **discrete choice models** and **network loading** ensures computational tractability while capturing nonlinear congestion thresholds ($f_b/C_b > 0.85$).

O奖提出多方法论框架，采用熵权聚合，Wardrop均衡，基于模糊综合评价模型量化多维度影响，兼顾交通流量、通勤效率和健康成本

Optimization and Evaluation of Baltimore's Transportation System Based on Network Models and Clustering

Summary

Transportation systems play a crucial role in the economy and residents' daily lives, yet they face complex challenges due to the diverse needs of various stakeholders. These challenges often result in parts of the system benefiting specific groups while impeding others' mobility. Baltimore, in particular, is struggling with aging infrastructure and limited transportation options, further exacerbated by the collapse of the Francis Scott Key Bridge, which has intensified traffic congestion. To address these issues, the city is focusing on repairing infrastructure, expanding public transit, and improving connections to ports and airports, with an aim to enhance traffic flow and foster economic growth. Additionally, with national funding, Baltimore plans to remove highways that divide neighborhoods, reconnect communities, and improve public housing and green spaces to promote sustainable development.

In this study, we developed a model to analyze and optimize Baltimore's transportation network. The model represents intersections as nodes and road segments as edges, differentiating between one-way and two-way roads. High-traffic intersections and primary roads are visually emphasized to provide a comprehensive foundation for system optimization. A key objective was to assess the impact of the Francis Scott Key Bridge collapse on the city's transportation system. Using traffic flow data and the Verhulst model, we predicted baseline traffic for 2025 assuming the bridge had not collapsed, enabling us to compare traffic flow changes before and after the incident.

We also employed traffic flow diffusion analysis to simulate how traffic spreads across the network, generating a citywide traffic flow map. Regional traffic analysis was conducted by cal-

F奖通过网络模型，交通流扩散分析交通网络，基于AHP筛选最佳方案，但较为依赖精确数据

Reimagining Baltimore's Flow: Enhancing Sustainable Mobility for Community Well-being

Summary

Baltimore is the largest city in Maryland, USA, and one of the important transportation hubs in the northeastern United States. However, its transportation system is plagued by problems such as fragile infrastructure and limited transportation options, which urgently need to be improved. We analyzed the influences of three projects on its transportation system, hoping to enhance the quality of life for residents.

Firstly, we analyzed the demands of **six** types of **stakeholders** and established a **Multilayer-Zoned Transportation Network** (MZTN) for vehicles, dividing the traffic system into **walk, bus, and drive** layers. This is conducive to mapping the travel demands of stakeholders onto the map. For question one, we established the **Tree-Based Traffic Allocation Model** to calculate the **traffic volume** before and after the bridge collapse and its reconstruction. By calculating the **traffic saturation**, it was known that the bridge collapse added approximately **30000 traffic flow** to two tunnels and one interstate highway. For different stakeholders, the **zonal Monte Carlo** method was employed to calculate the travel cost. Among them, the travel cost of **tourists** increased the most, reaching **48.9%**.

For question two, we proposed a selective **bus lane project** to promote traffic equality. Bus lanes will be established on roads that have **no less than three** lanes and where the percent of **households with no vehicle available** exceeds the average. When analyzing the influence, we calculate the travel time of private cars and buses through the **BPR (Bureau of Public Roads)** Function, obtaining that this project will reduce the travel time of city residents and suburban residents by **10.1%** and **3.1%** respectively, while increasing the travel time of tourists by **23.3%**. Particularly, for calculating the waiting time for buses, a probability model of **geometric distribution** was established. Implementing this project can reduce the waiting time, which meets the punctuality requirements of commuters.

For question three, we recommended the **Red Link project**, which can improve **transportation, economy, environment** and other aspects. Firstly, the accessible area of the Red

M奖建立多层分区交通网络，使用 BPR 函数与几何概率模型计算通勤时间，但依赖大量独立分析，忽视了地形和气候影响

Flow and Flux: The Pulse of Baltimore's Urban Pulse

Summary

Baltimore faces challenges related to aging infrastructure and limited transportation options, which negatively impact the daily lives of its residents. Recently, the collapse of a major bridge in the city has affected various stakeholders to varying degrees. Therefore, **Baltimore urgently needs appropriate projects to improve its transportation network**. This paper constructs three different models to assess the impact of various transportation projects on different groups in Baltimore.

First, to evaluate the impact of the bridge collapse, we assessed the importance of transportation nodes and selected several representative nodes to build a transportation network model. Using K-means clustering analysis, we examined the traffic flow distribution at these nodes and found that, following the collapse of the bridge, the traffic pressure on alternative river-crossing routes significantly increased. Additionally, the traffic pressure on several key nodes in the city also rose. Based on this finding, **the reconstruction of the Francis Scott Key Bridge was deemed necessary**.

Next, we studied Baltimore's bus network. Using the K-means algorithm, we divided all bus stops in Baltimore into nine regions based on their geographic locations. By integrating a traffic flow distribution model, we developed a formula to calculate the commuting time between regions, leading to the determination of the average commuting time for each region. We also proposed a public project to establish a metro trunk line running through the city and used the model to calculate changes in average commuting times for residents in different regions after the project's implementation. The results showed that **the metro trunk line project could effectively reduce commuting times across all regions, providing significant benefits to most stakeholders**.

Additionally, to improve commuting for Baltimore residents, we proposed adding new bus stops in residential areas and high-traffic zones. We extracted and visualized these areas, calculated distances between nodes using the Haversine formula, and merged nearby nodes. Applying K-means clustering, we analyzed residential areas and constructed an undirected graph to compute shortest path distances from each area to the nearest high-traffic node. Based on this, we identified

H奖通过K-均值聚类分析交通流量，基于最短路径分析确定新公交站点，但建模过程中信息损失较多，评价指标主观因素高

A Gift for Baltimore Residents: An Optimized Transportation System

Summary

The transportation system can either facilitate or hinder urban development and residents' lives. This paper addresses issues such as the collapse of the Francis Scott Key Bridge in Baltimore, the impact of I-95 highways that divide communities, and the underdevelopment of public transportation systems. **A network flow model** based on **edge sets, node sets**, and **event flows** is established. Using algorithms such as **A*, genetic algorithms, and dimensional analysis**, the transportation system is simulated and optimized.

Problem 1: To simulate the changes in the transportation system before and after the collapse of the Francis Scott Key Bridge, **a network flow model and a benefit model** are first developed. The model is based on graph theory with edge sets, node sets, and event flows. Data processing and longitude-latitude models are used to simulate the transportation system. **Static network search algorithms** and **A*** are then applied to calculate the shortest path, reflecting factors such as travel cost and time. Furthermore, **a benefit model** is established through weighting and dimensional analysis, successfully quantifying the overall benefit for the same group and comparing stakeholders' interests.

Problem 2: For the local optimization issue, Solution 1 is proposed. Starting with **the static network model**, additional node sets are introduced to complete **the network flow construction**. The dynamic simulation of the transportation system is achieved by considering node delay times. Noticing that I-95 isolates certain communities, a special segment of I-95 is converted into an ordinary road to better serve residents. The dynamic network simulation and benefit model are used to analyze the different impacts of Solution 1 on various stakeholders.

S奖基于图论开发网络流模型模拟交通系统，应用 A* 算法和遗传算法，但模型过于简化，文章排版存在较大问题

论文分评

上海交通大學
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PART FOUR

亮点分析——O奖

- **整体评估框架**：模型通过一个集成的网络框架，平衡了结构复杂性与动态交通均衡原则，对灾后交通韧性提供整体性、系统性的分析。

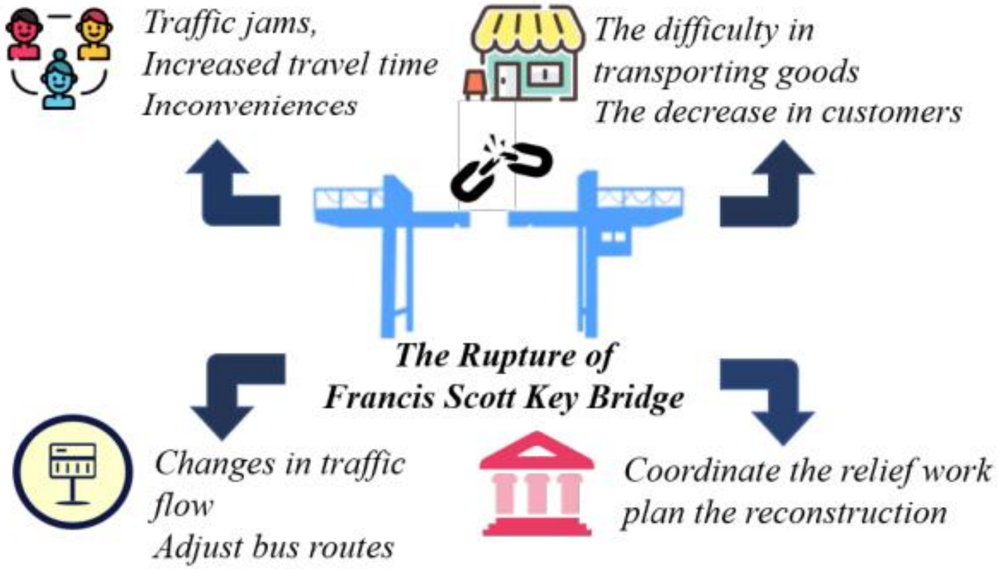


Figure 6: Impact of Francis Scott Key Bridge Collapse

亮点分析——O奖

- **多维评价体系**： 引入了**模糊综合评价模型 (Fuzzy Comprehensive Evaluation Model)**，将流量重分配、绕行负担 、环境外部性等多个维度整合进一个统一的评估体系。
- **方法融合提升精度**： 通过结合 **熵权法 (Entropy Weight Method)** 和 **Wardrop均衡 (Wardrop User Equilibrium)**两种核心方法，提升了预测结果的准确性，并经过了实证数据验证。

缺点分析——O奖

- **网络动态性考虑不足**：模型将交通网络视为静态（固定节点与边），忽略了**时变拥堵**和**季节性波动**，可能影响其在实时响应(real-time response)分析中的准确性。

亮点分析——F奖

- **方法先进且实用**：
  采用**弗胡斯特模型 (Verhulst Model)** 与**交通流扩散分析 (Traffic Flow Diffusion Analysis)** 提升预测准确性。
  结合**迪杰斯特拉算法 (Dijkstra's Algorithm)** 与**模拟退火算法 (Simulated Annealing)** 进行地铁线路优化，提供可操作的工程解决方案。

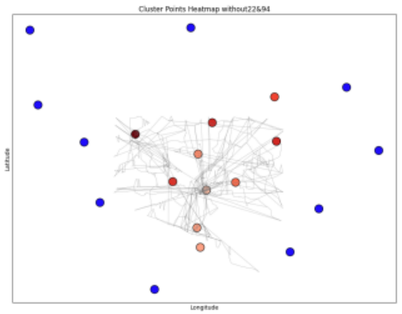


Figure 10: Key Point Inconvenience Heatmap

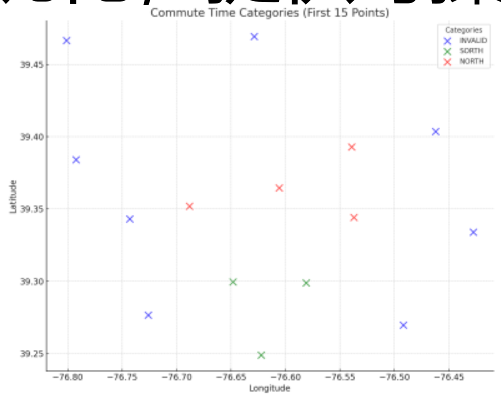


Figure 11: Effective Key Point Clustering Distribution

亮点分析——F奖

- **多维系统整合**：模型全面整合了**交通流分析**、**区域影响与利益相关者视角**，构建了多维度评估框架。
- **因素涵盖全面**：模型纳入**通勤者行为模式** 与**公交系统分析**等现实因素，增强了模型应对复杂现实场景的能力。

缺点分析——F奖

- **数据依赖性强**：模型精度高度依赖准确、一致的交通数据，实际应用中可能存在**数据可获性 (Data Availability)** 与**数据一致性 (Data Consistency)** 的挑战。
- **存在必要简化**：为保持实用性，模型采用了如**区域聚类假设 (Regional Clustering Assumption)** 等简化方法。虽然当前合理，但可能影响微观层面的精确度，未来需通过更细化数据或模型迭代进行完善。

亮点分析——M奖

- **多层分区网络结构**：模型采用**多层分区交通网络**，能够同步讨论不同交通模式与出行需求，网络结构紧凑且信息承载量高。

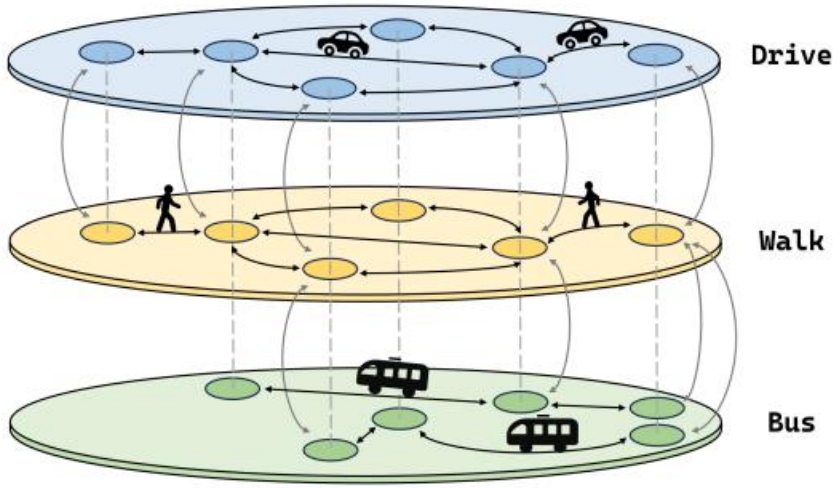


Figure 3: Transportation network

亮点分析——M奖

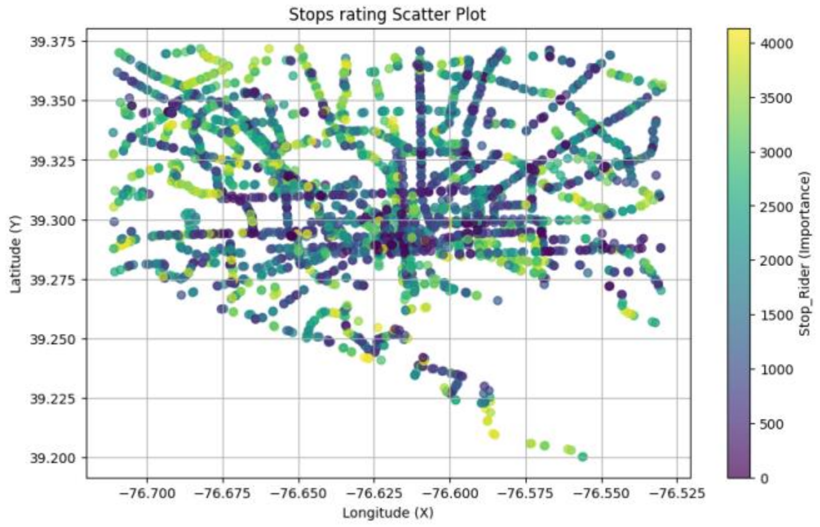
- **跨学科方法迁移应用**：创新性地将**引力模型 (Gravitational Model)** 应用于区域关联分析。将**数字结构**应用于交通流划分，增强了模型的创新性与实用性。
- **影响量化分析**：在分析不同利益相关者影响时，模型实现了多数影响的**量化分析 (Quantified Analysis)**，使结果更为直观且具有显著意义。
- **等待时间建模创新**：运用**几何分布概率模型 (Geometric Distribution Probability Model)** 分析公交等待时间，所得结论具有实践指导意义。

缺点分析——M奖

- **分区精度不足**：为简化计算，交通网络的**分区划分**较为粗糙，可能导致对不同出行需求的影响分析不够精确。
- **缺乏统一分析方法**：针对不同项目带来的深层影响，缺乏对利益相关者影响的**统一分析方法**，需对各项目进行具体分析。
- **环境因素考虑欠缺**：项目未纳入**地形影响**和**天气条件影响**，这些因素在本项目中被视为可忽略。

亮点分析——H奖

- **普适性强**：研究采用**K均值聚类分析 (K-means Clustering Analysis)** 对城市进行区域划分，有效降低了模型复杂度。这种数据驱动的分区方法兼具普适性与灵活性，可广泛适用于其他城市。



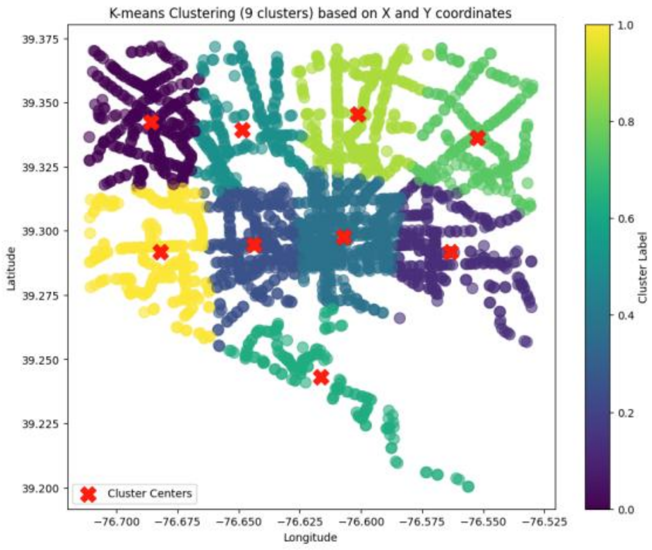


Figure 8: K-means Clustering(9 clusters)

亮点分析——H奖

- **动态响应能力**：模型能够动态展示事件发生时的相关数值变化，便于分析交通事件对不同区域和时段的影响。该特性使模型能反映交通系统的**实时响应 (Real-Time Response)**，有助于更精细地分析事件的动态演变过程。
- **强鲁棒性与稳定性**：模型对单一数值的波动不敏感，整体结果保持稳定。这表明模型具有较强的**鲁棒性**，能有效处理参数变化并减弱数据波动的影响，从而提升了模型的可靠性。

缺点分析——H奖

- **网络简化导致的信息损失**：在构建交通网络模型时，为简化分析过程而省略了部分细节与数据。这种简化可能导致一些复杂但重要的**交通模式**和**网络特性**信息丢失，可能影响模型的全面性与准确性。
- **时间模型缺乏物理意义**：在第二部分的通勤时间计算中，部分参数的选择基于人为假设。虽然这些假设有助于揭示区域间通勤时间的相对关系与差异，但所估计的平均通勤时间可能不具备实际的物理意义，限制了模型在特定具体场景下的适用性。

优点分析——S奖

- **模型构建数据驱动且实现全局优化**：基于**实际交通网络情况**与**真实数据**构建了网络流模型，并通过创新性结合**A*算法**与**遗传算法**，实现了**全局仿真**与**全局优化**，具有良好的模拟与描述效果。
- **多利益方考量与创新量化方法**：原创性地引入了**事件流集**概念，通过事件流的叠加构成整体道路交通系统的具体情况，并通过对权重的搜索实现了**满意度量化**，在全局优化中兼顾了各方利益平衡。
- **数据详实且可视化程度高**：研究收集并处理了大量数据，通过精确的**可视化**手段对网络流进行了模拟，增强了结果的表现力与可理解性。

优点分析——S奖

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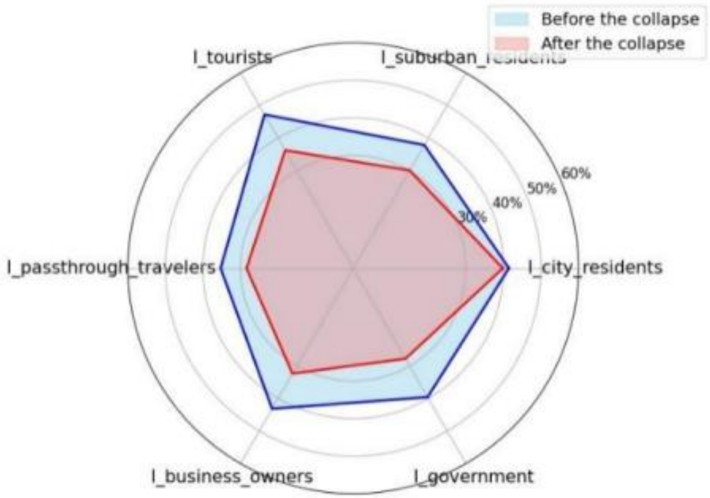

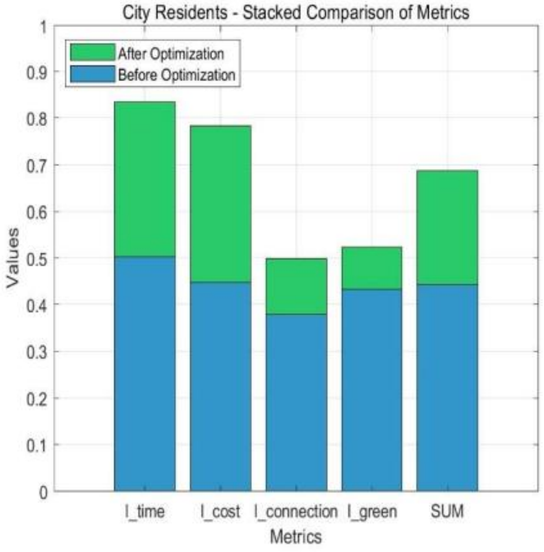

缺点分析：S奖

- **假设过于理想化**：模型忽略了摩托车与步行的区别，并假设“任何交通参与者均能在不变的路况下保持恒定速度”，该假设在实践中并非总是成立。
- **模型复杂度高**：本文提出的**网络模型**与**优化算法**较为复杂，涉及**多参数**与**多计算方法**。这可能导致模型的计算成本较高，尤其在进行大规模数据处理时，**计算时间**与**资源消耗**可能成为瓶颈。
- **参数估计存在不确定性**：模型中的部分参数，如**交通流量**、**延误系数**等，需通过数据估计与经验获得。这些参数的估计可能存在误差，进而影响模型的**精确性**与**可靠性**。

谢谢倾听

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